Beechcraft King Air B200 / 200 (B200)

2× Pratt & Whitney PT6A-42, 850 shp (PT6A-41 on the 200) · 2 crew + 6 to 13 seats · Typical B200 (PT6A-42, 6.5 psi cabin, hydraulic gear) AFM procedures, 12,500 lb; 200 / early B200 differences (750°C ITT, electric gear) noted in parentheses. Written for either the analogue panel or Pro Line 21.

V-speeds & limits

V1 / Vr	**94 KIAS (12,500 lb, flaps UP)**
V2	**approx 103 KIAS**
Vmca	**86 KIAS**
Vsse	**104 KIAS (intentional one-engine-inop)**
Vx / Vxse	**100 / 115 KIAS**
Vy / Vyse	**125 / 121 KIAS (blue line 121)**
Best glide	**approx 135 KIAS (both props feathered)**
Vfe	**200 KIAS (APPROACH) / 157 KIAS (DOWN)**
Vlo	**182 KIAS extend / 163 KIAS retract**
Vle	**182 KIAS**
Va	**181 KIAS (12,500 lb)**
Vmo	**259 KIAS**
Vs / Vso	**99 / 75 KIAS (12,500 lb)**
Vref	**approx 103 KIAS (flaps DOWN, max landing weight)**
Min icing speed	**140 KIAS (sustained icing)**

Normal procedures

Before Start

- Preflight inspection — **COMPLETE (covers, tie-downs, chocks removed; control locks REMOVED)**
- Weight & balance / takeoff data — **COMPUTED (flap setting, V1/Vr 94, V2 103, Vyse 121, field length)**
- Passenger briefing — **COMPLETE (belts, airstair door, emergency exit, oxygen masks, no smoking)**
- Seats, belts, shoulder harnesses — **ADJUST & LOCK**
- Cabin door / cargo door — **CLOSED, LOCKED (handle stowed, CABIN DOOR annunciator out)**
- Parking brake — **SET**
- Landing gear control — **DOWN**
- Power levers — **IDLE**
- Propeller levers — **FULL FORWARD (FEATHER permitted for cold-weather start)**
- Condition levers — **FUEL CUTOFF**
- Autofeather — **OFF**
- Avionics master / inverters / generators — **OFF**
- Bleed air valves — **OPEN**
- Oxygen — **CHECK (supply pressure, duration for passengers, crew masks)**
- Battery — **ON – voltage CHECK (approx 24 V; 23 V min for battery start)**
- Annunciators — **PRESS TO TEST (all lights and master warning / caution)**
- Fuel quantity — **CHECK main and aux (both sides)**
- Firewall shutoff valves — **OPEN (guards down)**
- Aux transfer — **AUTO (both)**
- Crossfeed — **AUTO (centre / OFF if no auto crossfeed)**
- Standby pumps — **ON (L / R FUEL PRESS lights OUT)**
- Beacon — **ON**

Engine Start

▶ Power source for start?

Battery:

Start the right engine first (airstair / cabin door side kept clear). Battery start: 23 V minimum.

External power:

- External power ..... **CONNECTED (EXT PWR annunciator) – switch ON, bus voltage approx 28 V**

Propeller area ..... **CLEAR (both sides)**

Ignition and engine start switch ..... **ON (START) – starter limits 40 sec ON / 60 sec OFF**

Oil pressure ..... **RISING**

N1 (Ng) ..... **12% MINIMUM, stable**

Condition lever ..... **LOW IDLE**

ITT / N1 ..... **LIGHT-OFF within 10 sec, N1 accelerating; MONITOR ITT (1000°C max, 5 sec)**

Abort start if no light-off within 10 sec, ITT approaches 1000°C, or N1 hangs: condition lever FUEL CUTOFF, continue motoring until ITT falls, then starter OFF.

Ignition and engine start switch ..... **OFF at approx 50% N1**

N1 – low idle ..... **approx 52–58% (stable)**

Oil pressure ..... **CHECK (60 psi min at idle; 100–135 psi normal)**

Propeller ..... **UNFEATHERED (prop lever full forward)**

Generator ..... **RESET, then ON (loadmeter / voltage check)**

▶ Second engine start method?

Cross-generator:

- Condition lever (running engine) ..... **HIGH IDLE (approx 70% N1)**
- Generator load ..... **BELOW 50% before engaging starter**

External power:

- External power ..... **ON until second engine running**

Second engine ..... **START – repeat sequence (N1 12% min, LOW IDLE, ITT 1000°C max, starter OFF 50%)**

Second generator ..... **RESET, then ON**

External power ..... **OFF & DISCONNECTED (if used; EXT PWR light out)**

Condition levers ..... **LOW IDLE (HIGH IDLE if bleed air / icing ops required)**

Standby pumps ..... **OFF (FUEL PRESS lights remain out)**

Inverter ..... **No. 1 ON – INVERTER light OUT; select No. 2 – CHECK; leave one ON**

Avionics master ..... **ON**

Engine instruments / annunciators ..... **CHECK both engines (ITT, torque, N1, oil pressure / temperature, fuel flow; no red lights)**

Before Taxi

Flight instruments / EFIS / avionics ..... **CHECK & SET (AHRS aligned on Pro Line 21; no flags; altimeters set; FMS loaded)**

Autopilot / electric trim ..... **TEST (trim both halves of split switch, disconnect, AP disengage) – OFF**

Pitot heat / stall warning heat / fuel vent heat ..... **ON momentarily – CHECK (loadmeter rise), then AS REQUIRED**

Propeller deice ..... **AUTO – ammeter 14–18 A; MANUAL inner / outer CHECK; then AS REQUIRED**

Surface deice (boots) ..... **SINGLE cycle – WING and TAIL DEICE lights, pressure, boots inflate / deflate**

Windshield anti-ice ..... **NORMAL – CHECK; HI in flight only**

Brake deice ..... **CHECK (if fitted) – OFF; do not use above 15°C**

Engine anti-ice (ice vanes) ..... **AS REQUIRED (ON below 5°C in visible moisture; ENG ANTI-ICE lights)**

Pressurisation ..... **TEST (condition levers HIGH IDLE, bleed air OPEN, switch to TEST – cabin pressurises, rate indicates descent; return to PRESS)**

Flaps ..... **SET for takeoff (UP or APPROACH) – indicator CHECK**

Trims ..... **SET for takeoff (elevator per loading; rudder / aileron neutral)**

Brakes ..... **CHECK on initial roll (power levers in BETA for speed control)**

Flight instruments ..... **CHECK in turns (HSI / compass, turn coordinator)**

Before Takeoff

Parking brake	**SET**
Flight controls	**FREE & CORRECT**
Overspeed governors	**TEST – hold PROP GOV TEST, advance power levers: props stabilise 1830–1910 RPM; release**
Primary governors	**EXERCISE – power to approx 1800 RPM, prop levers aft to detent (RPM decreases), then FULL FORWARD**
Rudder boost	**ON – TEST: one power lever up to approx 1000 ft-lb, other idle; pedal on the high-power side moves forward; repeat other side**
Autofeather	**TEST – hold TEST, power levers approx 500 ft-lb: retard one lever – opposite light OUT at approx 400 ft-lb, both OUT and prop feathers at approx 200 ft-lb; repeat other side**
Autofeather	**ARM (lights illuminate above approx 90% N1 on takeoff roll)**
Engine instruments	**CHECK (oil temp 10°C min, oil pressure 100–135 psi, ITT, N1, fuel flow)**
Fuel panel	**CHECK (firewall valves OPEN, aux transfer AUTO, crossfeed AUTO, standby pumps OFF / as required, NO FUEL TRANSFER out)**
Electrical	**CHECK (both generators ON, loads balanced, inverter ON, voltage 28 V)**
Pressurisation	**Bleed air valves OPEN, switch PRESS, controller SET (cruise altitude + 500 ft, rate approx 500 fpm)**
Propeller levers	**FULL FORWARD (2000 RPM)**
Propeller sync / yaw damp	**OFF**
Condition levers	**LOW IDLE (HIGH IDLE per operator SOP / icing / bleed air required)**
Flaps / trims	**SET & INDICATING (flaps UP or APPROACH)**
Ice protection	**AS REQUIRED (engine anti-ice ON below 5°C in visible moisture – reduced torque available; pitot heat ON)**
Lights	**AS REQUIRED**
Transponder / TCAS	**ALT / TA-RA**
Annunciators / doors	**CHECK (no red / amber; CABIN DOOR light out)**
Takeoff briefing	**COMPLETE (abort below V1 94; after V1: autofeather, gear UP, V2 103 to obstacle height, 121 KIAS; Vmca 86; 2230 ft-lb / 800°C limits)**

After Takeoff / Climb

Landing gear	**UP (positive rate, below 163 KIAS) – 3 green out, handle light out**
Flaps	**UP (clear of obstacles, above 121 KIAS)**
Autofeather	**ARMED (both lights ON)**
Climb power	**SET (torque per chart, ITT 800°C max – 750°C on PT6A-41, 1900–2000 RPM)**
Airspeed	**140 KIAS (normal climb) / 160 KIAS (cruise climb); 125 KIAS best rate**
Yaw damp	**ON (required above 17,000 ft)**
Propeller sync	**ON**
Pressurisation	**CHECK (cabin climbing per rate, differential increasing, 6.5 psi max)**
Engine anti-ice / ice protection	**AS REQUIRED**
Engine instruments / lights	**CHECK both engines; landing / taxi lights OFF**

Cruise

Cruise power	**SET (torque per chart, 1700–1800 RPM, ITT within limits)**
Propeller sync	**ON (props matched)**
Autofeather	**AS REQUIRED (OFF in cruise is common; must be ARMED for takeoff and landing)**
Fuel	**CHECK quantity main / aux, NO FUEL TRANSFER out, imbalance within 1000 lb; crossfeed as required**
Pressurisation	**CHECK (cabin altitude, rate, differential 6.5 psi max)**
Oxygen	**CHECK (pressure, masks available)**
Engine instruments	**CHECK (oil 100–135 psi / 0–99°C, ITT, N1 101.5% max)**
Ice protection	**AS REQUIRED (140 KIAS min in icing)**

Descent

Altimeters	**SET**
Pressurisation controller	**SET (destination field elevation + 500 ft; rate as required)**
Windshield anti-ice	**NORMAL (as required below 10,000 ft)**
Ice protection / engine anti-ice	**AS REQUIRED**
Fuel	**CHECK (crossfeed AUTO / OFF, standby pumps as required, aux transfer AUTO)**
Approach briefing	**COMPLETE (Vref approx 103, flap setting, single-engine go-around point)**
Seat belts / passenger briefing	**SECURE**

Before Landing

Autofeather	**ARM**
Pressurisation	**CHECK (cabin altitude at or above field elevation; differential approaching zero)**
Flaps	**APPROACH (below 200 KIAS)**
Landing gear	**DOWN (below 182 KIAS) – 3 GREEN, handle light out (HYD FLUID LOW out)**
Propeller sync / yaw damp	**OFF**
Condition levers	**LOW IDLE (HIGH IDLE per operator SOP / icing)**
Propeller levers	**FULL FORWARD (2000 RPM) on final**
Flaps	**DOWN (below 157 KIAS) when landing assured**
Airspeed	**Vref approx 103 KIAS (flaps DOWN); add approx 10–15 KIAS flaps UP / APPROACH**
Landing lights	**ON**
Landing gear	**3 GREEN – CONFIRMED on final**

After Landing

Power levers	**BETA / REVERSE as required – out of reverse by 40 KIAS**
Flaps	**UP**
Trims	**SET for takeoff**
Autofeather	**OFF**
Ice protection	**OFF (pitot / stall / vent heat, prop deice, windshield heat, brake deice as required)**
Engine anti-ice (ice vanes)	**AS REQUIRED (ON for ground operation below 5°C in visible moisture)**
Lights / transponder	**AS REQUIRED / STBY**
Condition levers	**LOW IDLE (HIGH IDLE if air conditioning / bleed air required)**

Shutdown

Parking brake	**SET**
Avionics master	**OFF**
Inverter	**OFF**
Ice and rain protection	**OFF**
Air conditioning / electrical load / standby pumps	**OFF**
ITT	**STABILISED – 1 minute at LOW IDLE**
Condition levers	**FUEL CUTOFF**
Propeller levers	**FEATHER**
ITT / N1	**MONITOR during run-down (no ITT rise)**
Exterior lights / beacon / oxygen	**OFF**
Battery	**OFF (after propellers stopped)**
Control locks	**INSTALL**
Chocks / covers / tie-downs	**AS REQUIRED**

Abnormal procedures

Generator Failure (L / R DC GEN annunciator)

DC GEN light, loadmeter zero

Affected generator switch	**OFF, then RESET, then ON**

▶ Generator restored?

Yes:

Loadmeters / voltage	**MONITOR both**

No – one generator failed:

Affected generator switch	**OFF**
Electrical load	**REDUCE below 100% on remaining generator**
Air conditioning / brake deice / windshield HI	**CONSIDER OFF**
Deferred: One generator inoperative – electrical load limited, land as soon as practical.	**NOTED**

No – both generators failed:

Both generator switches	**OFF**
Non-essential electrical load	**OFF (battery only – approx 30 min endurance)**
Avionics	**ESSENTIAL only (one nav / com, one inverter)**
Landing gear	**PLAN early extension while battery power remains; manual extension available**
Deferred: Both generators failed – battery only; extend gear early, flaps may be slow, land as soon as possible.	**NOTED**

Inverter Failure (INVERTER annunciator)

Inverter switch ..... **SELECT other inverter (No. 1 / No. 2)**

▶ INVERTER light out?

Yes:

AC instruments ..... **CHECK (torque meters, AC-powered avionics)**

Deferred: One inverter inoperative – single AC source, maintenance required. ..... **NOTED**

No – both failed:

Inverter ..... **OFF**

Power setting ..... **USE ITT, N1 and fuel flow – torque meters inoperative (analogue panel)**

AC-powered avionics / instruments ..... **NOTE inoperative; use standby instruments**

Deferred: Both inverters failed – torque indication / AC avionics inoperative, set power by ITT and N1. ..... **NOTED**

Low Oil Pressure / Chip Detector

Oil pressure below 100 psi above idle, OIL PRESS or CHIP DETECT annunciator

▶ Indication?

Oil pressure low:

Oil pressure / temperature ..... **CROSS-CHECK (60–100 psi at idle only; below 60 psi unsafe)**

▶ Pressure below 60 psi or temperature rising?

Yes:

Affected engine power ..... **REDUCE to minimum required**

Engine ..... **SHUT DOWN & FEATHER before seizure if terrain / altitude allow (Engine Failure in Flight)**

Deferred: Low oil pressure – precautionary shutdown, single-engine landing, land as soon as possible. ..... **NOTED**

No – 60–100 psi:

Power ..... **REDUCE; monitor pressure and temperature**

Deferred: Oil pressure in caution range – land as soon as practical, maintenance check. ..... **NOTED**

CHIP DETECT light:

Engine instruments (affected) ..... **MONITOR oil pressure / temperature, vibration**

Power ..... **REDUCE to minimum practical**

Engine ..... **SHUT DOWN & FEATHER if any other abnormal indication**

Deferred: Chip detector – land as soon as practical, engine inspection required. ..... **NOTED**

Fuel Pressure Low / Crossfeed / No Fuel Transfer

L / R FUEL PRESS or NO FUEL TRANSFER annunciator

▶ Annunciator?

FUEL PRESS (one side):

Standby pump (affected side) ..... **ON**

▶ FUEL PRESS light out?

Yes:

Standby pump ..... **LEAVE ON**

Deferred: Engine-driven boost pump failed – standby pump on; operation without boost pump limited to 10 hours cumulative. ..... **NOTED**

No:

Crossfeed ..... **SELECT toward affected engine (FUEL CROSSFEED light); standby pump supplying side ON**

Fuel quantity ..... **MONITOR both sides (both engines on one side's fuel)**

Firewall shutoff valve (affected) ..... **CHECK OPEN**

Deferred: Crossfeed in use – fuel from one side only, monitor quantity; land as soon as practical. ..... **NOTED**

NO FUEL TRANSFER:

Aux transfer switch (affected) ..... **OVERRIDE**

▶ Light out / aux quantity decreasing?

Yes:

Aux transfer ..... **LEAVE in OVERRIDE**

Deferred: Aux transfer in override – maintenance check required. ..... **NOTED**

No:

Aux fuel (affected side) ..... **CONSIDER UNUSABLE – replan fuel on main tanks, crossfeed to balance**

Deferred: Aux fuel not transferring – reduced usable fuel, imbalance within 1000 lb. ..... **NOTED**

Autofeather Failure / Not Armed

Autofeather lights not illuminated with power set, or AUTOFEATHER OFF annunciator

Autofeather switch ..... **CHECK ARM**

Power levers ..... **ABOVE approx 90% N1 (arming requires both power levers forward)**

▶ Both autofeather lights ON?

Yes:

System ..... **ARMED – continue**

No:

Autofeather switch ..... **OFF**

Autofeather must be armed for takeoff and landing. On the ground: do not take off. In flight: brief manual identify / verify / feather for an engine failure.

Deferred: Autofeather inoperative – manual feathering required if an engine fails; approach briefed accordingly. ..... **NOTED**

Rudder Boost Failure (RUDDER BOOST OFF annunciator)

Rudder boost switch ..... **OFF, then ON**

Bleed air valves ..... **CHECK OPEN (rudder boost uses bleed air pressure differential)**

▶ Rudder boost restored?

Yes:

System ..... **ON – continue**

No:

Rudder boost switch ..... **OFF**

Deferred: Rudder boost inoperative – expect full rudder force required on an engine failure; Vmca unchanged. ..... **NOTED**

Uncommanded rudder input:

Rudder boost switch ..... **OFF**

Rudder trim ..... **ADJUST for straight flight**

Deferred: Rudder boost off after uncommanded input – full rudder forces on an engine failure. ..... **NOTED**

Propeller Overspeed

Prop RPM above 2000 and not responding to the prop lever

Affected power lever ..... **REDUCE**

Affected propeller lever ..... **AFT toward minimum RPM / FEATHER detent**

Airspeed ..... **REDUCE**

▶ RPM controllable at or below 2000 (overspeed governor holds approx 2080)?

Yes:

Power ..... **SET to keep RPM within limits (2000 max; 2200 transient 5 sec)**

Propeller sync ..... **OFF**

Deferred: Primary governor fault – RPM controlled with power lever and airspeed, land as soon as practical. ..... **NOTED**

No – RPM above 2200 or uncontrollable:

Affected engine ..... **SHUT DOWN & FEATHER (Engine Failure in Flight)**

Deferred: Engine shut down for prop overspeed – single-engine landing, land as soon as possible. ..... **NOTED**

Bleed Air Failure (L / R BL AIR FAIL annunciator)

Bleed air line rupture / leak sensed

Bleed air valve (affected side) ..... **INSTR & ENVIR OFF**

Brake deice ..... **OFF**

Pressurisation ..... **MONITOR (single bleed source; cabin may climb at high altitude / low power)**

Condition levers ..... **HIGH IDLE if required in descent to maintain cabin**

▶ Cabin altitude maintained?

Yes:

Deferred: One bleed air source off – reduced pressurisation and rudder boost / surface deice capacity, maintenance required. ..... **NOTED**

No:

Altitude ..... **DESCEND to hold cabin below 10,000 ft**

Deferred: Bleed air failure – limited pressurisation, lower cruise altitude, maintenance required. ..... **NOTED**

Pressurisation Controller Failure / Cabin Not Following

Cabin altitude or rate not following controller; differential abnormal

Cabin pressurisation switch ..... **PRESS**

Bleed air valves ..... **OPEN**

Controller ..... **RESELECT cabin altitude / rate; cycle rate knob**

Cabin differential ..... **CHECK – 6.5 psi max (safety valve relieves above; 6.0 psi on the 200)**

▶ Cabin response?

Cabin following again:

Pressurisation ..... **MONITOR closely for remainder of flight**

Deferred: Pressurisation controller intermittent – maintenance check. ..... **NOTED**

Cabin climbing:

Altitude ..... **DESCEND to keep cabin below 10,000 ft; oxygen if cabin above 10,000 ft**

Use Pressurisation Loss / Rapid Decompression if the cabin continues to climb.

Deferred: Pressurisation controller failed – unpressurised or limited altitude, oxygen as required. ..... **NOTED**

Differential too high:

Altitude ..... **DESCEND – reduce differential**

Condition levers ..... **LOW IDLE; one bleed air valve ENVIR OFF to reduce inflow**

Before landing ..... **CABIN PRESSURISATION switch DUMP below 10,000 ft so cabin is depressurised at touchdown**

Deferred: Outflow valve fault – depressurise with DUMP before landing; do not open the door until differential is zero. ..... **NOTED**

Icing Conditions / Ice Accumulation

Engine anti-ice (ice vanes) ..... **ON (below 5°C in visible moisture) – ENG ANTI-ICE lights; expect lower torque**

Pitot heat / stall warning heat / fuel vent heat ..... **ON**

Propeller deice ..... **AUTO (ammeter 14–18 A); MANUAL if AUTO fails**

Windshield anti-ice ..... **NORMAL (HI if required in flight)**

Surface deice (boots) ..... **SINGLE cycle at first sign of ice – repeat as required**

Brake deice ..... **ON before landing if required (OFF within 10 min after gear retraction)**

Airspeed ..... **140 KIAS MINIMUM in sustained icing**

Autopilot ..... **DISCONNECT periodically – check for trim / control changes**

Exit icing ..... **CLIMB / DESCEND / TURN as required**

▶ Residual ice on airframe for landing?

No:

Ice protection ..... **OFF when clear**

Yes:

Flaps ..... **APPROACH only for landing**

Approach speed ..... **INCREASE approx 10–15 KIAS; no abrupt pitch changes**

Engine anti-ice ..... **LEAVE ON until shutdown**

Deferred: Residual airframe ice – APPROACH flaps, Vref +10–15 KIAS, longer landing distance. ..... **NOTED**

Pitot / Static Heat Failure

L / R PITOT HEAT annunciator with switch ON, or erratic airspeed in icing

Pitot heat switches ..... **CHECK ON (annunciator indicates heater failed or switch OFF)**

Pitot heat circuit breakers ..... **CHECK / RESET once**

▶ Annunciator out?

Yes:

Flight ..... **CONTINUE, monitor**

No:

Airspeed indications ..... **CROSS-CHECK pilot vs copilot vs GPS ground speed; use the unaffected side (ADC reversion on Pro Line 21)**

Icing conditions ..... **AVOID / EXIT**

Pilot's emergency static source ..... **SELECT if static blockage suspected (altimeter / VSI erratic)**

Deferred: Pitot heat failed one side – use opposite airspeed source; avoid icing; approach by attitude and power if unreliable. ..... **NOTED**

HYD FLUID LOW / Landing Gear Unsafe

HYD FLUID LOW annunciator, gear handle red light stays on, or fewer than 3 green

Airspeed ..... **BELOW 182 KIAS**

Landing gear control ..... **CONFIRM position selected; indicator bulbs CHECK (press to test / swap)**

Landing gear relay circuit breaker ..... **CHECK IN**

▶ Situation?

HYD FLUID LOW, gear up:

Landing gear ..... **PLAN manual extension (Landing Gear Manual Extension) – do not rely on normal extension**

Deferred: Hydraulic fluid low – manual gear extension, no retraction after extension. ..... **NOTED**

HYD FLUID LOW, gear down:

Landing gear ..... **LEAVE DOWN – do not retract**

Airspeed ..... **182 KIAS maximum**

Deferred: Gear down for remainder of flight – 182 KIAS max, increased fuel burn. ..... **NOTED**

Gear unsafe after selection:

Landing gear control ..... **RECYCLE once (UP then DOWN, below 163 KIAS)**

Gear horn / lights ..... **CHECK – landing gear warning horn with power levers retarded**

▶ 3 green?

Yes:

Deferred: Gear indication fault – do not retract after landing, maintenance inspection required. ..... **NOTED**

No:

Use Landing Gear Manual Extension; if still unsafe, plan a gear-unsafe landing (longest runway, emergency services).

Emergency procedures

Engine Failure During Takeoff (below V1)

On the runway, below 94 KIAS / decision speed

Power levers (memory) ..... **IDLE (both)**

Brakes (memory) ..... **MAXIMUM – stop straight ahead**

Power levers (memory) ..... **GROUND FINE / REVERSE as required (operating engine – anticipate yaw)**

Condition levers ..... **FUEL CUTOFF (if stopping / fire suspected)**

Firewall shutoff valves ..... **CLOSED (if fire suspected)**

Battery / generators ..... **OFF after stop (if evacuating)**

Below Vmca (86 KIAS) directional control cannot be maintained with full power on one engine. Below V1 always stop.

Engine Failure After V1 / Liftoff

At or above V1 (94 KIAS), committed to fly

LAND AS SOON AS POSSIBLE

Power (memory) ..... **MAXIMUM ALLOWABLE (2230 ft-lb / 800°C – 750°C PT6A-41)**

Airspeed (memory) ..... **TAKEOFF SPEED to V2 103 KIAS until clear of obstacles**

Landing gear (memory) ..... **UP (positive rate)**

Propeller (memory) ..... **VERIFY AUTOFEATHERED – if not: identify, verify, prop lever FEATHER**

Airspeed (memory) ..... **121 KIAS (Vyse) once clear of obstacles**

Flaps (memory) ..... **UP (above 121 KIAS, obstacles cleared)**

Bank / rudder ..... **3–5° toward operating engine, ball half out (rudder boost assists)**

Trim ..... **RUDDER & AILERON for straight flight**

Climb ..... **121 KIAS to safe altitude, then Engine Failure in Flight – securing items**

Deferred: One engine inoperative after takeoff – single-engine approach and landing, no go-around once flaps DOWN. ..... **NOTED**

Engine Failure in Flight / Engine Securing

Loss of power on one engine at altitude

LAND AS SOON AS POSSIBLE

Power (memory)	**AS REQUIRED (max 2230 ft-lb / 800°C)**
Airspeed (memory)	**121 KIAS (Vyse) minimum**
Inoperative engine (memory)	**IDENTIFY – dead foot, dead engine (torque / N1 / ITT)**
Inoperative engine power lever (memory)	**VERIFY – retard, no change**
Inoperative engine condition lever (memory)	**FUEL CUTOFF**
Inoperative engine propeller lever (memory)	**FEATHER (confirm autofeather / manual)**
Firewall shutoff valve (inop side) (memory)	**CLOSED**
Generator (inop side)	**OFF**
Autofeather	**OFF**
Propeller sync	**OFF**
Bleed air valve (inop side)	**ENVIR OFF (INSTR & ENVIR OFF if contamination suspected)**
Standby pump (inop side)	**OFF**
Electrical load	**MONITOR (single generator, 100% max)**
Rudder boost	**ON – CHECK operating**
Trim	**RUDDER & AILERON (3–5° bank toward operating engine)**
Crossfeed	**AS REQUIRED (standby pump on supplying side ON)**
Pressurisation	**MONITOR (single-engine bleed; cabin altitude may rise)**

▶ Cause known and no damage – attempt air start?

Yes – air start:

Use the Air Start procedure. If unsuccessful, leave the engine secured.

No – engine secured:

ATC	**ADVISE – nearest suitable airport, drift-down altitude check**
Deferred: One engine inoperative – single-engine approach and landing, crossfeed as required, no go-around once flaps DOWN.	**NOTED**

Air Start

Engine secured, cause known, no fire or mechanical damage

Cabin temperature mode ..... **OFF**

Electrical load ..... **REDUCE (single generator)**

Condition lever ..... **FUEL CUTOFF**

Firewall shutoff valve ..... **OPEN**

Standby pump ..... **ON**

Power lever ..... **IDLE**

▶ Air start method?

Starter-assisted (any altitude):

- Propeller lever ..... **FEATHER**
- Ignition and engine start switch ..... **ON (START) – observe starter limits**
- N1 ..... **12% MINIMUM**

Windmilling (below 20,000 ft, 140 KIAS min):

- Propeller lever ..... **FULL FORWARD (unfeather – prop windmills)**
- N1 ..... **12% MINIMUM (use starter if less)**
- Ignition and engine start switch ..... **ON**

Condition lever ..... **LOW IDLE**

ITT ..... **LIGHT-OFF within 10 sec; MONITOR (1000°C max)**

Ignition and engine start switch ..... **OFF at approx 50% N1**

▶ Engine restarted?

Yes:

- Oil pressure ..... **CHECK**
- Propeller lever ..... **FULL FORWARD / AS REQUIRED (match RPM)**
- Generator ..... **RESET, then ON**
- Bleed air valve ..... **OPEN**
- Standby pump ..... **OFF (FUEL PRESS out)**
- Power ..... **AS REQUIRED after warm-up (ITT / torque in limits)**

Deferred: Engine restarted in flight – land as soon as practical, maintenance investigation required. ..... **NOTED**

No:

- Condition lever ..... **FUEL CUTOFF**
- Ignition and engine start switch ..... **OFF**
- Propeller lever ..... **FEATHER**
- Firewall shutoff valve ..... **CLOSED**
- Standby pump ..... **OFF**

Do not attempt further starts. Complete the Engine Securing items.

Deferred: One engine inoperative – single-engine approach and landing. ..... **NOTED**

Engine Fire in Flight

FIRE annunciator / visual fire

LAND IMMEDIATELY

Condition lever (affected) (memory) ..... **FUEL CUTOFF**

Propeller lever (affected) (memory) ..... **FEATHER**

Firewall shutoff valve (affected) (memory) ..... **CLOSED**

Fire extinguisher (if fitted) (memory) ..... **PUSH TO ARM / DISCHARGE**

Generator (affected) ..... **OFF**

Bleed air valve (affected) ..... **INSTR & ENVIR OFF**

Standby pump (affected) ..... **OFF**

Autofeather / propeller sync ..... **OFF**

Airspeed ..... **121 KIAS minimum (increase if fire persists)**

Electrical load / crossfeed ..... **AS REQUIRED**

Do not attempt a restart. Land at the nearest suitable airport; if fire persists or structure is involved, land immediately on any suitable surface.

Deferred: Engine fire – single-engine landing, evacuation on landing, fire services required. ..... **NOTED**

Engine Fire on the Ground / During Start

- Condition lever (memory) ..... **FUEL CUTOFF**
- Firewall shutoff valve (memory) ..... **CLOSED**
- Ignition and engine start switch (memory) ..... **CONTINUE MOTORING (starter) to purge fuel / lower ITT**
- Fire extinguisher (if fitted) ..... **PUSH TO ARM / DISCHARGE**
- Standby pumps ..... **OFF**

▶ Fire out?

Yes:

- Ignition and engine start switch ..... **OFF (observe starter limits)**
- Other engine ..... **SHUT DOWN – condition lever FUEL CUTOFF, props FEATHER**
- Battery / generators ..... **OFF**

Do not restart. Maintenance inspection required.

No:

- Ignition and engine start switch ..... **OFF**
- Other engine ..... **SHUT DOWN – condition lever FUEL CUTOFF**
- Battery / generators ..... **OFF**
- Aircraft ..... **EVACUATE (away from the fire side), ground extinguisher**

Emergency Descent

- Power levers (memory) ..... **IDLE**
- Propeller levers (memory) ..... **FULL FORWARD**
- Flaps (memory) ..... **APPROACH (below 200 KIAS)**
- Landing gear (memory) ..... **DOWN (below 182 KIAS)**
- Airspeed (memory) ..... **181 KIAS (Va)**
- Oxygen ..... **AS REQUIRED (crew masks ON above 10,000 ft cabin)**
- ATC ..... **ADVISE – emergency descent**
- Level-off ..... **10,000 ft or MEA – restore power smoothly, gear up below 163 KIAS if continuing**

Rapid Decompression / Cabin Altitude Warning

CABIN ALT / ALT WARN annunciator (cabin above 12,500 ft) or loud decompression

LAND AS SOON AS PRACTICAL

- Crew oxygen masks (memory) ..... **DON – 100%; mic switch OXY MASK**
- Passenger oxygen (memory) ..... **ON (passenger manual override PULL if masks have not dropped)**
- Emergency descent (memory) ..... **INITIATE to 10,000 ft or MEA**
- Passenger / crew communication ..... **ESTABLISH**
- Cabin pressurisation ..... **CHECK (bleed air valves OPEN, switch PRESS, controller set) – see Pressurisation Loss**
- ATC ..... **ADVISE**

▶ Cabin pressurisation recovered?

Yes:

- Cabin altitude ..... **MONITOR; stay at an altitude the cabin can hold**
- Deferred: Pressurisation fault – fly at low altitude, oxygen used, maintenance inspection. ..... **NOTED**

No:

- Altitude ..... **REMAIN at or below 10,000 ft (oxygen supply as required)**
- Deferred: Unpressurised – remain at or below 10,000 ft, oxygen quantity limited. ..... **NOTED**

Pressurisation Loss / Cabin Altitude Climbing

Cabin altitude rising uncommanded, CABIN ALT warning not yet active

Cabin pressurisation switch ..... **PRESS (not DUMP / TEST)**

Bleed air valves ..... **OPEN (both)**

Pressurisation controller ..... **CHECK setting (cabin altitude selected below flight altitude; rate increase)**

Condition levers ..... **HIGH IDLE (if N1 low in descent – more bleed air)**

Cabin door / BL AIR FAIL annunciators ..... **CHECK**

Oxygen ..... **CREW MASKS ON if cabin above 10,000 ft**

▶ Cabin altitude controlled?

Yes:

Flight ..... **CONTINUE at an altitude the cabin maintains below 10,000 ft**

Deferred: Pressurisation degraded – limit cruise altitude, maintenance inspection. ..... **NOTED**

No:

Descent ..... **INITIATE – before cabin reaches 12,500 ft; Rapid Decompression procedure if warning activates**

Deferred: Unpressurised – remain at or below 10,000 ft. ..... **NOTED**

Smoke / Fume Elimination / Electrical Fire

LAND AS SOON AS POSSIBLE

Crew oxygen masks (memory) ..... **DON – 100% / EMER; mic switch OXY MASK**

Smoke goggles (memory) ..... **ON (if fitted)**

▶ Source of smoke?

Electrical:

Affected equipment / circuit ..... **OFF – circuit breaker PULL (if identified)**

▶ Source identified and smoke clearing?

Yes:

Deferred: Electrical fire – affected circuit isolated, equipment unavailable. ..... **NOTED**

No:

Generators ..... **OFF (battery only – essential loads via triple-fed bus)**

Avionics master / non-essential load ..... **OFF**

Hand fire extinguisher ..... **USE as required**

Electrical equipment ..... **RESTORE one item at a time, essential only, checking for smoke**

Landing gear is electro-hydraulic: plan a manual gear extension and a flaps-up landing if electrical power cannot be restored.

Deferred: Electrical fire – battery only / minimum equipment, manual gear extension and flaps-up landing possible. ..... **NOTED**

Environmental / bleed air:

Bleed air valves ..... **ENVIR OFF one at a time to identify, then INSTR & ENVIR OFF on affected side**

Cabin temperature mode ..... **OFF**

Altitude ..... **DESCEND to maintain cabin below 10,000 ft (single bleed source)**

Deferred: Bleed air source isolated – reduced pressurisation, lower cruise altitude. ..... **NOTED**

Unknown:

Both procedures ..... **APPLY – electrical isolation and bleed air isolation**

Smoke / fume evacuation:

Descent ..... **TO 10,000 ft or below if practicable**

Cabin pressurisation switch ..... **DUMP (below 10,000 ft) to ventilate**

Cabin air / fresh air ..... **MAXIMUM (bleed air OPEN if not the source)**

Passenger oxygen ..... **CONSIDER (does not protect from smoke – masks are diluter type)**

Landing Gear Manual Extension

Gear does not extend normally / HYD FLUID LOW

Airspeed ..... **BELOW 182 KIAS (reduce to approx 130 KIAS to ease pumping)**

Landing gear relay circuit breaker ..... **PULL (right subpanel)**

Landing gear control ..... **DOWN**

▶ Gear system?

Hydraulic (B200):

Emergency extension handle ..... **UNSTOW (floor, left of pilot seat) – PUMP up and down until 3 GREEN, then until resistance**

Emergency extension handle ..... **LEAVE in position – do not stow until gear inspected**

Electric (200 / early B200):

Hand crank ..... **ENGAGE (floor behind pedestal) – TURN CLOCKWISE to stop (approx 50 turns) – 3 GREEN**

Hand crank ..... **DISENGAGE & STOW**

Landing gear indicator ..... **3 GREEN, handle light out**

Do not reset the landing gear circuit breaker or move the gear control in flight after a manual extension. Leave the gear down.

▶ 3 green obtained?

Yes:

Deferred: Gear extended manually – do not retract; gear system inspection required. ..... **NOTED**

No:

Landing ..... **GEAR UNSAFE – longest runway, emergency services, flaps DOWN, minimum speed, fuel cutoff before touchdown**

Deferred: Gear unsafe landing – emergency services, evacuate after stopping. ..... **NOTED**

Single-Engine Approach, Landing & Go-Around

One engine secured and feathered

Inoperative engine ..... **SECURED (condition lever CUTOFF, prop FEATHER, firewall valve CLOSED, generator OFF)**

Autofeather ..... **OFF (inoperative with one engine)**

Rudder boost ..... **ON**

Crossfeed ..... **AS REQUIRED (balance for approach; standby pump supplying side ON)**

Electrical load ..... **WITHIN single generator limit (100%)**

Pressurisation ..... **SET destination + 500 ft (cabin may not hold – oxygen as required)**

Airspeed ..... **121 KIAS (Vyse) until landing assured**

Flaps ..... **APPROACH (below 200 KIAS)**

Landing gear ..... **DOWN when landing assured (below 182 KIAS) – 3 GREEN**

Propeller lever (operating) ..... **FULL FORWARD**

Flaps ..... **DOWN only when landing assured (short final, below 157 KIAS)**

Airspeed ..... **Vref approx 103–110 KIAS on short final**

Power lever (operating) ..... **REDUCE progressively – anticipate yaw change; rudder trim toward neutral**

▶ Go-around required?

No:

Touchdown ..... **NORMAL – BETA / REVERSE on operating engine with caution (asymmetric), brakes as required**

Yes – above decision height:

Power (operating) ..... **MAXIMUM ALLOWABLE (2230 ft-lb / 800°C)**

Flaps ..... **APPROACH (from DOWN)**

Landing gear ..... **UP (positive climb)**

Airspeed ..... **121 KIAS (Vyse) minimum; bank 3–5° toward operating engine**

Flaps ..... **UP (clear of obstacles, above 121 KIAS)**

A single-engine go-around with flaps DOWN or below approx 200 ft AGL should not be attempted – single-engine climb performance is marginal at high weight / altitude / temperature.